

Study Overview

Brief Summary

Patients who have histologically confirmed metastatic or unresectable (not amenable to curative therapy) breast cancer may be screened for eligibility. All patients must have HER2 negative breast cancer with the identified PIK3CA mutation and received at least one line of endocrine therapy. The study will consist of a screening phase, a treatment phase, and a post-treatment phase which includes safety, efficacy, and follow-up. The treatment phase will include taking alpelisib daily in combination with continued use of either Fulvestrant or Aromatase Inhibitor per standard of care until disease progression or unacceptable toxicity.

Official Title

A Phase II, Single Arm, Non-randomized Study of Alpelisib (BYL719) in Combination With Continued Endocrine Therapy Following Progression on Endocrine Therapy in Hormone Receptor Positive, HER2 Negative, PIK3CA Mutant Metastatic Breast Cancer

Contacts and Locations

This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record.

This study has 4 locations

United States

Illinois Locations

Chicago, Illinois, United States, 60612

University of Illinois Cancer Center

Nebraska Locations

Omaha, Nebraska, United States, 68198

University of Nebraska Medical Center

Pennsylvania Locations

Hershey, Pennsylvania, United States, 17033

Penn State Cancer Institute

Wisconsin Locations

Madison, Wisconsin, United States, 53705

University of Wisconsin

Participation Criteria

Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read Learn About Studies.

Eligibility Criteria

Description

Eligibility Criteria Individuals from populations who are underrepresented in clinical research (e.g., racial and ethnic minorities, women, individuals from rural/frontier communities, older individuals) will be enrolled with a goal of ensuring that all eligible patients are given the opportunity to participate in novel clinical trials and that research findings can be generalizable to the entire population.

Inclusion Criteria

Written informed consent and HIPAA authorization for release of personal health information. NOTE: HIPAA authorization may be included in the informed consent or obtained separately.

Age $\geq$ 18 years at the time of consent.

ECOG Performance Status of 0-2 within 28 days prior to registration.

Men and postmenopausal female patients. Premenopausal patients (age 18 or older) who have been rendered postmenopausal will also be included. Postmenopausal is defined as:

Age $\geq$ 55 years and one year or more of amenorrhea.

Age < 55 years and one year or more of amenorrhea, with estradiol < 20 pg/ml

Age < 55 years with prior hysterectomy but intact ovaries, with estradiol < 20 pg/ml

Prior bilateral oophorectomy

NOTE: Women who do not fit the criteria for being postmenopausal as above are deemed premenopausal. Premenopausal patients (age 18 or older) who can be rendered postmenopausal will also be eligible. Methods eligible for rendering premenopausal patients postmenopausal include:

Ongoing treatment with luteinizing hormone-releasing hormone (LHRH) agonist, with treatment starting at least 4 weeks prior to randomization and with estradiol < 20 pg/ml. LHRH agonist must be administered within 7 days of scheduled administration date during the length of the clinical trial.

Histologically confirmed metastatic or unresectable (not amenable to curative therapy) breast cancer.

Has confirmed hormone receptor positivity with ER $\geq$ 1% and/or PR $\geq$ 1%. Preferred receptor testing is obtained from a metastatic site, but can be from the primary breast or axilla biopsy as long as this is most recent biopsy.

Has confirmed HER2 negative breast cancer. HER2 negative or non-amplified is determined by the current ASCO-CAP criteria which are as follows: HER2 testing by IHC as 0 or 1+. Or negative by in situ hybridization (FISH/CISH/SISH) defined as Her2/CEP17 ratio <2 and for single probe assessment a HER2 copy number <6). Preferred receptor testing is obtained from a metastatic site, but can be from the primary breast or axilla biopsy as long as this is most recent biopsy.

PIK3CA mutation identified via local testing from tumor tissue or blood.

Has either measurable disease, i.e. at least one measurable lesion as per RECIST 1.1 criteria OR if no measurable disease is present, then at least one predominantly lytic bone lesion must be present, within 28 days prior to registration. Patients with no measurable disease and only one predominantly lytic bone lesion that has been previously irradiated are eligible if there is documented evidence of disease progression of the bone lesion after irradiation.

Has received at least one line of therapy with an endocrine therapy or endocrine therapy combination with other agents in the metastatic setting.

Must have received endocrine therapy (aromatase inhibitor or fulvestrant as single agent or in combination with other agents) as last line of therapy with progressive disease, as determined by treating physician.

No more than two prior lines of endocrine therapy or endocrine therapy combinations in the metastatic setting. Transition from one nonsteroidal aromatase inhibitor to a second nonsteroidal aromatase inhibitor due to tolerance will only count as one line of endocrine therapy. Combination therapy will be considered one line even if one of the agents is discontinued and the other continued.

Patient must have received treatment with a CDK4/6 inhibitor for breast cancer in the metastatic setting. CDK4/6 inhibitor discontinuation due to intolerance or reasons other than progression is allowed.

Demonstrate adequate organ function as defined in the protocol; all screening labs to be obtained within 28 days prior to registration.

Men who are not surgically or medically sterile must agree to use an acceptable method of contraception. Acceptable forms of contraception include castration, vasectomy, and condom/occlusive cap with spermicidal

foam/gel/film/cream/suppository. Male patients with female sexual partners who are pregnant, possibly pregnant, or who could become pregnant during the study must agree to use condoms at least one month after the last dose of study drug. Total abstinence for the same study period is an acceptable alternative.

Premenopausal females must agree to use an acceptable method of contraception or to abstain from sexual intercourse for the duration of their participation in the study and for 3 months after discontinuation of therapy. Adequate forms of contraception include: total abstinence, surgery intended to prevent pregnancy (defined as hysterectomy, bilateral oophorectomy or bilateral tubal ligation), non-hormonal IUD, or condom/occlusive cap with spermicidal foam/gel/film/cream/suppository).

As determined by the enrolling physician or protocol designee, ability of the

subject to understand and comply with study procedures for the entire length of the study

HIV-infected patients on effective anti-retroviral therapy with undetectable viral load within 6 months of registration are eligible for this trial.

Exclusion Criteria

Patients with prior chemotherapy for metastatic or advanced disease.

Active infection requiring systemic therapy.

Patients with a prior or concurrent malignancy whose natural history or treatment has the potential to interfere with the safety or efficacy assessment of the investigational regimen are not eligible for this trial.

Uncontrolled, active CNS metastases causing clinical symptoms or metastases that require therapeutic intervention, including leptomeningeal disease.

NOTE: Subjects who are symptomatic and have not undergone prior brain imaging must undergo a head computed tomography (CT) scan or brain magnetic resonance imaging (MRI) within 28 days prior to registration to exclude brain metastases.

Treatment with any investigational drug within 14 days prior to registration.

Radiotherapy to index lesion $\leq$ 4 weeks or limited field radiation to index lesion for palliation $\leq$ 2 weeks prior to randomization.

Established diagnosis of diabetes mellitus type I or persistent poorly controlled diabetes mellitus, with an uninterrupted hemoglobin A1c $> 8.0\%$ for 1 year or greater despite standard care. For patients with newly diagnosed diabetes mellitus without 1 year of hemoglobin A1c values, available hemoglobin A1c values cannot all be $> 8.0\%$.

Symptomatic visceral disease or any disease burden that makes the patient ineligible for endocrine therapy per the investigator's best judgment.

As determined by the enrolling physician or protocol designee, impairment of gastrointestinal function or disease that may significantly alter the absorption of the study drugs.

Currently documented clinically active pneumonitis. Patients could have received prior treatment for pneumonitis but pneumonitis must have clinically resolved and treatments for pneumonitis (e.g. steroids) must be completed prior to randomization.

Active cardiac disease, defined as any of the following within 6 months prior to the start of study treatment:

History of angina pectoris, coronary artery bypass graft, symptomatic pericarditis, or myocardial infarction.

History of documented congestive heart failure (New York Heart Association functional classification III-IV).

History of any cardiac arrhythmias, (e.g., ventricular tachycardia), complete left bundle branch block, high grade AV block (e.g., bifascicular block, Mobitz type II and third degree AV block), supraventricular, nodal arrhythmias, or conduction abnormality.

Systolic blood pressure (SBP) > 180 mmHg and/or Diastolic Blood Pressure (DBP) ≥ 100 mm Hg at screening. Initiation or adjustment of antihypertensive medication(s) is allowed prior to screening.

History of acute pancreatitis within 1 year of screening or a past medical history of chronic pancreatitis.

Prior treatment with PI3K, mTOR or AKT inhibitor in the metastatic setting.

History of chronic steroid use (defined as daily steroid use > 14 days) and requirement for continued chronic steroid use.

Study Plan

This section provides details of the study plan, including how the study is designed and what the study is measuring.

Expand all / Collapse all

How is the study designed?

Design Details

Primary Purpose : Treatment

Allocation : N/A

Interventional Model : Single Group Assignment

Interventional Model Description: Apolisib 300mg PO, Daily AND Aromatics Inhibitor(AI) or Fulvestrant 500mg IM per Standard of Care
Masking : None (Open Label)
Arms and Interventions

Participant Group/Arm Intervention/Treatment

Experimental: Apolisib + Aromatase Inhibitor or Fulvestrant

Subjects will be treated with Apolisib in combination with either an Aromatase Inhibitor or Fulvestrant per Standard of Care

Drug: Apolisib

Apolisib 300mg, PO, days 1-28 of each cycle.

Drug: Fulvestrant

Fulvestrant 500mg, IM, once monthly

Drug: Aromatase inhibitor

Aromatase Inhibitor, administered per standard of care

What is the study measuring?

Primary Outcome Measures

Outcome Measure	Measure Description	Time Frame
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Progression-Free Survival (PFS)	To estimate the progression-free survival (PFS) of apolisib with continued endocrine therapy (aromatase inhibitor or fulvestrant) following progression in patients with hormone receptor positive, HER2 negative, PIK3CA mutant metastatic breast cancer. PFS defined as time from D1 of treatment with apolisib with endocrine therapy.	From enrollment until the time of disease progression, up to 60 months
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Secondary Outcome Measures

Outcome Measure	Measure Description	Time Frame
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Objective Response Rate (ORR)	To estimate the objective response rate (ORR) of apolisib with continued endocrine therapy (aromatase inhibitor or fulvestrant) following progression in patients with hormone receptor positive, HER2 negative, PIK3CA mutant metastatic breast cancer. ORR will include confirmed complete response (CR) and confirmed partial response (PR) as determined by RECIST 1.1	From enrollment until the time of disease progression, up to 60 months
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Clinical Benefit Rate (CBR)	To estimate the clinical benefit rate (CBR) of apolisib with continued endocrine therapy (aromatase inhibitor or fulvestrant) following progression in patients with hormone receptor positive, HER2 negative, PIK3CA mutant metastatic breast cancer. CBR will include the sum of CR + PR + stable disease (SD) on treatment with apolisib with endocrine therapy.	From enrollment until the time of disease progression, up to 60 months
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Duration of Response (DOR)	To estimate the duration of response (DoR) of apolisib with continued endocrine therapy (aromatase inhibitor or fulvestrant) following progression in patients with hormone receptor positive, HER2 negative, PIK3CA mutant metastatic breast cancer. DOR is defined as the time that measurement criteria are met CR or PR (whichever status is recorded first) until the date that recurrent or progressive disease PD) is objectively documented.	From enrollment until the time of disease progression, up to 60 months
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Overall Survival (OS)	To estimate the overall survival (OS) of apolisib with continued endocrine therapy (aromatase inhibitor or fulvestrant) following progression in patients with hormone receptor positive, HER2 negative, PIK3CA mutant metastatic breast cancer. OS is defined as the time from treatment initiation with apolisib with endocrine therapy following progression until death as a result of any cause.	From enrollment until the time of death, up to 60 months
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Safety Profile of apolisib with continued endocrine therapy(aromatse inhibitor or fulvestrant)	To estimate grade 3-4 adverse events or events of any grade leading to dose adjustments of apolisib with continued endocrine therapy (aromatase inhibitor or fulvestrant) following progression in patients with hormone receptor positive, HER2 negative, PIK3CA mutant metastatic breast cancer. Grade 3-4 toxicities as defined by the NCI Common Terminology Criteria for Adverse Events (NCI CTCAE) v5 or events of any grade leading to dose adjustments.	From enrollment until 30 days after completion of study therapy or subject withdrawal, up to seven months
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Collaborators and Investigators

This is where you will find people and organizations involved with this study.

Sponsor

Marina N Sharifi

Collaborators

Novartis

University of Wisconsin, Madison

Investigators

Principal Investigator: Marina N Sharifi, MD, University of Wisconsin, Madison

Study Record Dates

These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.

Study Registration Dates

First Submitted

2021-02-17

First Submitted that Met QC Criteria

2021-02-17

First Posted

2021-02-21

Study Record Updates

Last Update Submitted that Met QC Criteria

2025-12-30

Last Update Posted

2026-01-02

Last Verified

2025-12